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Revolutionizing clinical laboratories: The impact of artificial intelligence in diagnostics and patient care

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ABSTRACT

Introduction: The integration of artificial intelligence (AI) is fundamentally transforming clinical laboratories, significantly improving diagnostic precision and operational effectiveness in the fields of pathology, microbiology, and biochemistry. This evolution holds great promise for advancing patient care and enhancing disease management strategies.

Methodology: A comprehensive literature review was performed using established databases, including Google Scholar, Embase, Science Direct, Scopus, and PubMed. Recent studies published between 2020 and 2024 were sourced using focused keywords related to AI's application in clinical laboratory settings. The inclusion criteria prioritized peer-reviewed articles that contributed to innovations in diagnostic methodologies and operational efficiency. A thematic analysis was conducted to collate findings regarding AI's impact across the preanalytical, analytical, and postanalytical phases of laboratory work.

Discussion: AI significantly enhances various laboratory processes, such as histopathology, immunohistochemistry, and microbiological diagnostics. Notable applications include workflow automation, detailed analysis of biomarker data, and real-time processing to facilitate clinical decision-making. However, the benefits of AI come with challenges, including concerns about data integrity, ethical implications, and potential biases in algorithms, requiring careful management as AI becomes more integrated into clinical practice.

Conclusion: The future of clinical laboratories is poised for increased automation and the incorporation of AI and IoT technologies. While these advancements offer the potential for improved healthcare outcomes through greater accuracy and efficiency, evolving ethical and legal frameworks are crucial to address issues related to data privacy and accountability of algorithms. Ongoing adaptation and exploration of AI applications are vital to fully harnessing its capabilities in diagnostics.

1. Introduction

Ensuring the "quadruple aim" of healthcare—enhancing population health, improving patient and carer experiences, and reducing escalating care costs—remains a significant challenge for global healthcare systems [1]. Artificial Intelligence (AI), a field focused on enabling computers to emulate human intelligence with minimal human intervention, is poised to revolutionize healthcare by integrating seamlessly with existing healthcare systems at a pivotal juncture [2]. This integration is being driven by the increasing availability of multimodal data, including genomic, clinical, demographic, economic, and phenotypic datasets. Additionally, advancements in mobile technology, the Internet of Things (IoT), computing power, and data security further accelerate the adoption of AI in healthcare delivery models [3].

AI technology in healthcare encompasses both machine learning (ML) and non-ML methodologies. Machine learning, a subset of AI, utilizes algorithms to analyze datasets for tasks such as detection and classification, enabling autonomous pattern recognition across diverse domains [4]. In contrast, non-ML approaches rely on deterministic models and traditional statistical methods, focusing on analysis and prediction without adaptive algorithms.

Laboratory medicine has long been a highly digitalized field, with automation, electronic reporting, and digital result transmission widely adopted since the advent of these technologies [5]. Clinical tests provide

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critical diagnostic information, and integrating AI into laboratories can significantly reduce the workload on clinical laboratory staff by automating the entire testing process [6]. Additionally, medical laboratories manage vast databases containing test results and quality control data, often within intricate quality management systems. Consequently, laboratory medicine has become a frontrunner in the digital transformation of the medical field. However, despite these advancements, many areas within laboratory medicine have yet to fully embrace recent developments in data science, including machine learning and artificial intelligence. The current momentum, however, suggests that the time is ripe for these innovations to take root [7].

The integration of AI into clinical laboratories presents an unprecedented opportunity to enhance efficiency, accuracy, and scalability in laboratory medicine. By leveraging AI-driven insights and automation, laboratories can optimize workflows, minimize human error, and bolster diagnostic support, aligning with healthcare's overarching goals of improving patient outcomes and reducing costs. To fully realize AI's potential, fostering interdisciplinary collaboration among Pathology, Microbiology, and Biochemistry will be crucial, propelling laboratory medicine into a new era of data-driven healthcare. This article explores AI's transformative role across these specialties, highlighting its significance, applications, and implications for the future of diagnostics.

2. Methodology

2.1. Literature search

We conducted a thorough literature search to locate pertinent research on the application of AI in clinical laboratories from a variety of fields, such as pathology, microbiology, and biochemistry. Reputable databases such as Google Scholar, Embase, Science Direct, Scopus, and PubMed were utilised for selecting the papers. The following keywords and phrases were used to gather recent, notable research published between 2020 and 2024: "artificial intelligence in clinical laboratories," "artificial intelligence in laboratory diagnosis," "AI in pathology," "AI in microbiology," and "AI in biochemistry."

2.2. Inclusion and exclusion criteria

Peer-reviewed publications that discussed AI developments in clinical laboratory settings for patient care, operational effectiveness, and diagnostic accuracy were the main emphasis of the inclusion criteria. To guarantee that the data is current, only research that has made substantial contributions to the area and recent advancements were included. Studies with outdated approaches or no clinical application were excluded based on exclusion criteria.

2.3. Data analysis and synthesis

The inclusion criteria focused primarily on peer-reviewed studies that addressed AI advancements in clinical laboratory settings for patient care, operational effectiveness, and diagnostic accuracy. Only studies that have made significant contributions to the field and the most recent advances were included to ensure that the data is accurate and up to date. According to the exclusion criteria, studies that used outdated approaches or had little clinical use were not included.

3. Discussion

3.1. Clinical pathology

Pathology aids as a foundation of modern medicine, offering vital insights into disease diagnosis, prognosis, and treatment planning. With the ever-increasing intricacy and volume of clinical data, traditional pathology methods face challenges in maintaining consistency, accuracy, and efficiency. Artificial Intelligence (AI) has emerged as a transformative tool, offering advanced solutions to these challenges by leveraging machine learning (ML), deep learning (DL), and other computational techniques. This review explores AI's impact on clinical pathology, focusing on its applications in histopathology, immunohistochemistry (IHC), cytology, image-based analysis, and chemotherapy response scoring.

AI's integration into pathology signifies a paradigm shift, automating labor-intensive processes, reducing variability, and enabling objective data interpretation. As AI technologies mature, their role in enhancing personalized medicine, streamlining workflows, and improving diagnostic precision becomes increasingly apparent. However, implementing AI in pathology is not without challenges, such as ensuring data quality, achieving algorithm generalization, and addressing ethical considerations. These aspects underscore the need for a balanced perspective when evaluating AI's potential in pathology.

3.1.1. AI in histopathology and immunohistochemistry (IHC)

Histopathology and IHC play pivotal roles in disease identification, particularly in oncology. Traditional methods rely on subjective tissue sample evaluation, which can be time-consuming and variable. AI, especially through machine learning and deep learning techniques, offers automation and refinement for these processes.

- **Tumor Detection and Classification**: AI models demonstrate promise in tumor detection and subtype classification. For example, Esteva et al. (2017) showed that deep learning algorithms can classify skin cancer types with accuracy comparable to that of expert dermatologists [8]. Another critical application of AI is in tumor grading, a process with significant implications for many tumor types. For example, prostate cancer grading has benefitted from AI-assisted techniques, as demonstrated by Egevad et al. (2020), which identified areas of grading difficulties and compared them with AI-assisted grading, highlighting its potential to enhance accuracy and consistency [9]. AI is also making strides in radiopathomics, the intersection of radiology and histopathology. For example, McGarry et al. (2018) demonstrated how radiologic findings could predict the location of clinically significant, high-grade prostate cancer by integrating radiopathomic maps of epithelium and lumen density [10].
- **Quantitative Analysis**: AI enables the quantitative analysis of histopathology images, such as assessing tumor cell density or invasion depth, which supports objective evaluations and personalized treatment planning [11].
- **Automation of Staining and Scoring in IHC**: AI has automated various aspects of IHC, traditionally assessed manually by pathologists.
  - **Staining Optimization**: AI algorithms enhance staining quality and consistency in IHC processes [12].
  - **Automated Scoring Systems**: AI systems measure biomarker intensity and distribution, automating IHC stain scoring. Yoo et al. (2017) reported that AI-driven automation minimizes interobserver variability, improving reliability [13].
  - **Image Interpretation**: AI supports the interpretation of complex IHC visuals, assisting with multiple biomarkers or spatial relationships within tissue [14].
- **Consistency in Scoring Systems**: Manual interpretations of staining intensity often vary among pathologists, impacting diagnoses such as estrogen receptor (ER) and progesterone receptor (PR) status. AI-based systems provide consistent scoring, reducing variability both between and within observers. This standardization supports more objective analyses of staining intensity and biomarker distribution [15,16].

Inconsistencies in immunohistochemistry, especially with ER and PR reporting, can significantly impact patient management. Advances in AI, combined with digital pathology and standardization initiatives, offer

promising solutions for improved reproducibility and reliability in IHC.

3.1.2. AI in cytology

AI is increasingly applied to Pap smear analysis, with the potential to improve diagnostic accuracy, streamline workloads, and enhance screening efficiency in cervical cancer detection.

- **Automated Detection and Classification**: AI systems, notably convolutional neural networks (CNNs), identify abnormal cells in Pap smears, distinguishing between normal and abnormal cells with high accuracy [17]. AI models further categorize cellular abnormalities as low-grade (LSIL) or high-grade squamous intraepithelial lesions (HSIL) based on cell morphology and nuclear atypia [18].
- **Reproducibility**: By standardizing Pap smear evaluations across laboratories, AI reduces variability and enhances result reproducibility.
- **Integration with Screening Programs**: Integrating AI into existing screening frameworks enhances efficiency and accessibility, broadening the reach of cervical cancer prevention efforts [19].

AI holds promise in improving the accuracy and efficiency of Pap smear assessments. However, challenges like data quality, model generalization, and ethical considerations must be addressed to maximize AI's potential in cervical cancer screening.

3.1.3. AI in image-based analysis

- **Automated Tumor Detection and Classification**: CNNs recognize malignancy patterns and features in histopathology images, enabling tumor detection without manual labeling [11].
- **Tumor Classification**: AI categorizes tumors into subtypes, such as breast cancer or glioblastoma, based on histological features, assisting in prognosis and treatment planning [20].
- **Cell Counting and Morphology**: AI automates cell counting and morphology assessments, measuring cellular characteristics and monitoring morphological changes associated with disease progression [17].
- **Staining Intensity and Pattern Analysis**: AI quantifies staining intensity and patterns, correlating these metrics with disease severity and prognosis [21].

3.1.4. Applications of AI in chemotherapy response scoring

- **Tumor Shrinkage Measurement**: AI algorithms analyze histopathology images to assess tumor shrinkage pre- and post-treatment, estimating treatment efficacy with precision [11].
- **Residual Tumor Detection**: AI detects residual tumor cells post-treatment by identifying subtle changes in tumor structure, indicating partial responses or residual disease [22].
- **Histological and Morphological Analysis**: AI detects treatment-induced cellular changes, such as necrosis, fibrosis, and apoptosis, critical indicators of treatment efficacy [17].
- **Biomarker Analysis**: AI assesses biomarkers linked to chemotherapy responses, analyzing patterns of immunohistochemical stains for markers like Ki-67 and PD-L1, aiding in evaluating treatment potential [21].

AI can optimize treatment protocols by predicting patient responses to specific chemotherapy based on histopathological profiles.

AI models face challenges in generalizing across diverse patient populations, clinical settings, and tumor types, requiring rigorous testing in varied environments. Integrating AI into clinical workflows requires careful assessment of usability and its impact on pathologists' responsibilities. Effective training and support are essential for seamless adoption [23]. Ongoing advances in deep learning techniques, including novel architectures and algorithms, may further enhance AI's accuracy and utility in chemotherapy response assessment [8].

Overall, AI integration within pathology can significantly enhance diagnostic accuracy, streamline processes, and support personalized treatment approaches. Applications span image analysis, IHC, and response scoring. However, to fully capitalize on AI's potential in pathology, challenges in data quality, algorithm generalization, and clinical integration must be addressed.

In conclusion, AI has emerged as a transformative force in clinical pathology, redefining traditional workflows and enabling precision medicine. Its applications in histopathology, IHC, cytology, image-based analysis, and chemotherapy response assessment demonstrate the breadth of its potential. By automating routine tasks, standardizing interpretations, and offering novel insights, AI enhances diagnostic accuracy and efficiency.

To fully realize AI's potential in pathology, challenges such as ensuring data quality, algorithm scalability, and ethical considerations must be addressed. Collaborative efforts between technologists, pathologists, and policymakers are essential to integrating AI seamlessly into clinical practice. With ongoing advancements, AI is set to revolutionize pathology, ultimately improving patient care and outcomes on a global scale.

3.2. Clinical microbiology

Clinical microbiology plays an essential role in the diagnosis of infectious diseases, guiding effective treatment, and monitoring antimicrobial resistance. Microbial diagnosis, a crucial element in healthcare, traditionally relies on culture, molecular analysis, and imaging to identify microorganisms. However, these conventional methods are labor-intensive, time-consuming, and prone to inaccuracies and often delay treatment, leading to challenges in effective diagnosis and treatment outcomes [24]. Such limitations have eventually led to the exploration of innovative technologies to enhance diagnostic capabilities. AI, has emerged as a pioneering tool, offering solutions to long-standing challenges in microbial diagnostics and paving the way for advanced, efficient, and accurate laboratory practices [24–26]. Beyond diagnostics, the transformative impact of AI also addresses challenges in resource-limited settings by providing cost-effective diagnostic tools and reducing dependency on extensive infrastructure, making it indispensable in clinical microbiology.

Sample collection and transport are critical to maintaining microorganism viability, yet improper handling or delays can compromise results, potentially leading to inaccurate diagnoses [25,26]. Additionally, culture and isolation methods, the foundation of traditional diagnosis, are time-intensive and may yield false negatives if cultivation conditions are inadequate. Antibiotic overuse further complicates the growth and isolation of certain organisms [27].

Traditional identification methods, such as biochemical tests and microscopy, may yield inaccurate results due to misleading morphological characteristics or atypical organism behaviors [28]. Bacterial susceptibility testing, while necessary, can also present challenges. Methods like disk diffusion require significant time and may not accurately predict the infecting strain's antibiotic response, often resulting in suboptimal treatment choices and associated complications [29]. The technical demands of these methods necessitate skilled personnel, specialized equipment, and substantial resources, all of which increase the risk of human error, particularly in resource-limited settings [30].

Delays in diagnosis can lead to uncontrolled disease progression and reliance on empirical therapy, often contributing to antibiotic resistance and misdiagnosis of emerging pathogens [31]. Cross-contamination and false positives add further complexity [32]. Traditional methods may also lack real-time data for effective epidemiological surveillance, and inconsistent procedures across laboratories can result in unreliable outcomes [32,33].

With advancements in microbiology, AI, particularly machine learning (ML), has emerged as a transformative tool, addressing

challenges in drug target prediction, infectious disease diagnosis, and understanding microbial resistance [34]. AI applications in clinical microbiology now include large datasets encompassing genomic information, metagenomics, mass spectra from cultured isolates, and digital imaging, all contributing to AI-driven diagnostic solutions. AI's integration has revolutionized the field of clinical microbiology by enhancing diagnostic precision and efficiency, as these capabilities significantly reduce the turnaround time for diagnostics, enabling targeted treatments within hours rather than days [35].

With millions worldwide affected by treatment-resistant infections each year, the rise of superbugs represents a major healthcare challenge in the era of antimicrobial resistance. Traditional methods require time-consuming cultivation and resistance testing of pathogens. In response, Pattern Bioscience has developed an innovative AI-powered assay that bypasses these steps, allowing healthcare providers to swiftly identify the most effective antibiotic for an infection. This rapid approach aims to curb the overuse of unnecessary or broad-spectrum antibiotics by enabling targeted therapy to begin in as little as four hours, a significant improvement over the two-to-four-day timeframe required for conventional culture-based testing [36].

AI's capabilities extend to virology, parasitology, mycology, and bacteriology, offering powerful tools for rapidly and accurately analyzing complex data [37,38]. For example, during the COVID-19 pandemic, Kaur et al. noted AI's ability to track and manage infected individuals efficiently. AI-based screening, analysis, and prediction methods outperformed non-AI-based methods in identifying COVID-19 symptoms, reducing time, costs, and human effort [39].

AI is also applied to diagnosing diseases like malaria, which typically require multiple diagnostic steps. Digital in-line holographic microscopy (DIHM), an AI-driven technology, effectively detects infected red blood cells in malaria patients, providing a cost-effective solution that reduces the need for extensive blood sample processing [35]. In public health, AI supports infectious disease management by aiding diagnosis, monitoring treatment, and enabling personalized care, as well as infection prevention by analyzing large datasets for outbreak patterns and strategy development [34,35,40]. During COVID-19, AI facilitated automatic virus spread predictions, infection tracking, and trend forecasting, which helped monitor remission probabilities [35].

3.2.1. *Challenges and ethical considerations in AI-driven microbial diagnostics*

While AI offers significant advantages, its application in microbial diagnostics raises ethical concerns around patient privacy, algorithmic bias, data security, transparency, and equitable treatment [41]. Compliance with legal and regulatory standards is essential for data handling, analysis, and sharing. As AI and ML become more integral to microbiology, discussions on stricter regulations around data privacy, ethics, and patient consent will likely increase. Establishing harmonized legal frameworks guided by FAIR principles (Findable, Accessible, Interoperable, and Reusable) can improve data visibility, usability, and interoperability [42].

AI's integration in healthcare policies and ethical frameworks promises transformative changes in microbiology and diagnostics. The "laboratory of the future" envisions heightened automation, robotic reliance, and AI and IoT connectivity, potentially enabling real-time personal health monitoring and disease detection. AI's role is anticipated to grow in areas like vaccine development, antimicrobial resistance tracking, and epidemiological surveillance, enhancing infectious disease management capabilities [43]. Key trends include improving data quality, developing interpretable AI models, addressing digitalization challenges, and navigating legal and ethical complexities. These developments have the potential to elevate microbiological diagnostics, infection prevention, and patient care, driving better outcomes in healthcare and public health [26].

In conclusion, AI is revolutionizing clinical microbiology by tackling challenges in diagnostics, antimicrobial resistance, and public health. Its ability to process complex data enhances accuracy, reduces delays, and facilitates informed decision-making. However, ethical concerns, data privacy, and transparency remain critical issues. As AI advances, its integration into microbiology holds great promise for improving diagnostics, treatments, and patient outcomes globally. The future depends on adopting these innovations while maintaining ethical and equitable practices.

3.3. *Clinical biochemistry*

Inclusion of artificial intelligence (AI) into biochemistry signifies a transformative advancement, bridging traditional biochemical practices with cutting-edge technology. AI has the potential to revolutionize biochemical diagnostics, laboratory workflows, and data interpretation by improving efficiency, accuracy, and predictive capabilities. Given the increasing complexity of data generated in clinical and research biochemistry, the adoption of AI tools is becoming indispensable.

The integration of AI into biochemistry is transforming the landscape of biochemical diagnostics and laboratory workflows. Leveraging machine learning and deep learning, AI enhances the accuracy, speed, and efficiency of diagnostics while streamlining laboratory operations across all phases of testing: preanalytic, analytic, and postanalytic processes [44,45].

3.3.1. *AI applications across testing phases*

3.3.1.1. Preanalytic phase. The preanalytic phase involves ordering tests, collecting patient samples, and preparing specimens for analysis. This stage, often occurring outside the lab's direct oversight, is prone to errors that can affect diagnostic accuracy. AI enhances oversight and error identification by automating workflows and improving communication between healthcare providers and laboratories, ultimately increasing testing reliability and minimizing preanalytic mistakes.

3.3.1.2. Analytic phase. AI is revolutionizing the analytic phase by processing complex biomarker data crucial for diagnosing diseases like cancer, cardiovascular conditions, and metabolic disorders. Traditional biomarker interpretation is time-intensive and susceptible to error due to data volume. AI-driven models can analyze this data in real time, detecting disease indicators—such as protein or genetic anomalies—earlier than clinical symptoms appear, thus enabling timely intervention and improving patient outcomes.

3.3.1.3. Postanalytic phase. In the postanalytic phase, AI improves the reporting process by facilitating automated result validation and interpretation, ensuring efficiency and reducing errors. By streamlining autoverification, AI helps to ensure that only validated, accurate results are reported into Laboratory Information Systems (LIS) and Electronic Health Records (EHR).

AI optimizes Clinical Biochemistry laboratory operations by enabling predictive staffing and resource management, analyzing real-time data to ensure appropriate personnel and reagent availability. Additionally, predictive maintenance algorithms monitor equipment performance, minimizing downtime and enhancing efficiency. Such AI applications, validated in other industries, hold significant promise for laboratory medicine, reducing operational disruptions and improving workflow management.

Challenges remain in maintaining AI accuracy. Issues such as quality control, specimen integrity, and interpretation require advanced AI models capable of nuanced decision-making. Furthermore, AI algorithms must mitigate biases and accommodate variability in regional clinical practices to ensure reliability.

3.3.2. Key applications

3.3.2.4. Automated interpretation of biomarker patterns for disease diagnosis and monitoring. AI's capacity to interpret complex biomarker patterns is transforming diagnostics for diseases like cardiovascular conditions and metabolic disorders [46–48]. By analyzing vast biomarker datasets, AI models, particularly neural networks, identify correlations that allow for early disease detection, even before clinical symptoms manifest. AI-driven analysis of protein levels, genetic expressions, and metabolite concentrations can reveal indicators of conditions like Alzheimer's, diabetes, and cancers, supporting disease monitoring and treatment evaluation [49–51].

3.3.2.5. Optimization of laboratory workflows. AI optimizes biochemical laboratory workflows by automating sample processing, data entry, and analysis. This automation reduces human intervention, minimizes errors, and improves lab throughput, crucial for swift diagnostic turnaround times. Additionally, AI enhances quality control by analyzing historical lab data to predict errors or equipment failures, maintaining consistent testing reliability [52–54].

3.3.2.6. Predictive analytics and decision support. AI's role in predictive analytics and clinical decision support integrates patient data with biochemical parameters to offer clinicians actionable insights [55]. For instance, AI can assess biomarker trends to predict disease recurrence in cancer patients or recommend personalized treatments by analyzing metabolic profiles. This AI-driven precision approach facilitates tailored treatment plans, aligning with the goals of precision medicine [56].

3.3.3. Ethical considerations

The ongoing integration of AI in biochemistry brings both significant opportunities and risks, necessitating a careful approach to development and adherence to ethical principles. Emerging ethical guidelines emphasize the importance of fairness, transparency, and accountability in AI applications within clinical biochemistry. Furthermore, ethical frameworks, tailored to pathology and laboratory medicine, underscore the necessity for data privacy, unbiased algorithms, and equitable patient treatment.

In general, the future of AI in biochemistry promises advanced automation, predictive analytics, and personalized patient care, with applications poised to further improve diagnostic accuracy and laboratory efficiency. As AI continues to shape the field, its ethical implementation will be paramount in ensuring equitable, high-quality healthcare outcomes.

3.3.4. Importance of AI in biochemistry

Artificial intelligence (AI) has emerged as a transformative tool in biochemistry, revolutionizing how data is processed, analyzed, and applied in both research and clinical settings. The field of biochemistry generates vast amounts of data, particularly in areas like biomarker discovery, metabolic profiling, and drug development. Traditional methods of data analysis are often time-intensive and prone to human error, limiting the ability to derive actionable insights quickly. AI addresses these challenges by automating complex analytical processes and uncovering patterns that may go unnoticed by human analysis.

In diagnostics, AI enhances the accuracy and speed of disease detection by analyzing intricate biomolecular data, such as protein structures, genetic variations, and metabolite levels. For instance, AI-powered models can predict disease states earlier than conventional diagnostic methods, enabling timely interventions and improving patient outcomes.

Moreover, AI optimizes laboratory workflows by automating processes in the preanalytic, analytic, and postanalytic phases [44,45]. This reduces human error, improves efficiency, and ensures the reliability of results. It also enables predictive analytics, allowing laboratories to anticipate equipment failures, reagent shortages, or personnel requirements, ensuring uninterrupted operations.

AI's role in precision medicine is another critical application. By integrating biochemical parameters with patient data, AI systems can recommend personalized treatments, monitor disease progression, and evaluate therapeutic responses. This aligns with the overarching goals of modern biochemistry to deliver tailored, patient-centric care.

Despite its potential, the adoption of AI in biochemistry must navigate challenges such as data privacy, algorithmic biases, and the need for standardized protocols. Nonetheless, its transformative potential makes it an indispensable tool in advancing the field, fostering innovation, and improving healthcare outcomes.

In conclusion, Artificial intelligence is redefining the field of biochemistry through automation, improved diagnostics, and predictive analytics. While challenges such as ethical considerations, algorithmic biases, and variability in clinical practices persist, the potential benefits of AI are undeniable. As AI continues to evolve, it is imperative for the biochemistry community to embrace its integration responsibly, ensuring adherence to ethical guidelines and a focus on patient-centered outcomes. Future research should address current limitations while exploring innovative applications to maximize the impact of AI in clinical and research biochemistry.

4. Future of AI in clinical laboratories

Clinical laboratories and diagnostic testing play a vital role in healthcare by enabling earlier, faster, and more accurate diagnoses that improve patient outcomes and can save lives. The advancements in diagnostic technologies have been transformative, and ongoing innovations promise to further enhance this field [56].

Future laboratories are expected to be increasingly automated, utilizing robotics, AI, and the IoT for improved connectivity and efficiency. AI has applications across various medical domains, extending beyond clinical and laboratory diagnostics to broader areas of patient care. However, the widespread adoption of AI in clinical laboratories requires careful consideration of legal accountability. Current uncertainties, lack of clarity and misunderstandings surrounding sensitive personal information, data processing, consent, transparency, and storage raise complex questions about the legal responsibility for AI-driven decisions in diagnostic settings [57].

Effectively integrating AI and ML in clinical laboratories demands substantial time and resources. Interpreting AI recommendations is often challenging, yet this investment can significantly enhance lab performance. For decision-makers, a practical approach involves identifying workflow obstacles and researching purpose-built AI solutions that align with the lab's needs. Providers experienced in clinical lab challenges can offer guidance on appropriate AI applications, ensuring targeted improvements [57,58].

Optimizing AI and ML use within laboratories is an ongoing process. Incremental adjustments and real-world insights often yield the most substantial improvements, establishing AI as an evolving asset for laboratory medicine [58].

5. Conclusion

AI is transforming clinical laboratories by improving diagnostic precision, speed, and efficiency in pathology, microbiology, and biochemistry. Through workflow automation, enhanced biomarker analysis, and personalized care, AI advances laboratory medicine. With robotics and IoT integration on the horizon, ethical and legal considerations remain critical. Ongoing adaptation holds promise for better healthcare outcomes via precision, efficiency, and diagnostic innovation.

Ethical approval

The authors confirm that this review was conducted in compliance with the ethical guidelines of K S Hegde Medical Academy (Nitte – Deemed to be University), principles outlines in the Declaration of Helsinki and standards set by the Committee on Publication Ethics (COPE). Because of the nature of this review, IEC (Institutional Ethics Committee) approval was not required.

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CRediT authorship contribution statement

Asem Ali Ashraf: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Srinidhi Rai:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Formal analysis, Data curation. **Sameeksha Alva:** Writing – review & editing, Writing – original draft, Validation, Software, Resources, Methodology, Formal analysis, Data curation. **Priya D. Alva:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Data curation. **Sriram Naresh:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Software, Methodology, Investigation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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